

CYCLISTIC BIKESHARE CASE STUDY

How Do Annual Members and Casual Riders Use Bikes Differently?

I. Introduction

Cyclistic is a Chicago-based bike-share program offering three pricing options, single-ride passes, full-day passes and annual memberships. Riders who purchase single-ride or full-day passes are classified as casual riders; riders with annual memberships are classified as members.

Lily Moreno, Director of Marketing, believes the company's future growth depends on maximizing annual memberships. Rather than building a campaign aimed at entirely new customers, she sees a stronger opportunity in converting casual riders, who are already using and familiar with Cyclistic, into annual members.

This report addresses the first of three questions guiding that strategy.

"How do annual members and casual riders use Cyclistic bikes differently?"

Assumptions

- Data-privacy restrictions prevent linking ride data to personally identifying rider information, so this analysis is limited to ride-level behavioral patterns (duration, timing, frequency) rather than demographics or individual rider history.
- Rides with a duration of zero (or negative) minutes or longer than 24 hours, are treated as data or system errors rather than genuine rider behavior, consistent with Divvy's own published policy of billing bikes not returned within 24 hours as lost or stolen.

II. Analysis and Findings

Data and Methodology

This analysis uses 12 months (Jan–Dec 2025) of public Divvy trip data, the real-world dataset standing in for Cyclistic's historical rides. The raw data totaled roughly 5.55 million rides across 12 monthly files. Data was combined and cleaned in Python (pandas), checking exact and ride-ID duplicates, dropping rows missing critical fields and calculating ride duration, day of week, hour, month and duration bucket as new fields.

Rides with non-positive or 24+ hour durations were removed (5,614 rows, well under 1% of the data). Findings were built and cross-validated across three tools, Python/pandas, Google BigQuery (SQL) and Tableau, to confirm consistency rather than relying on a single tool's output.

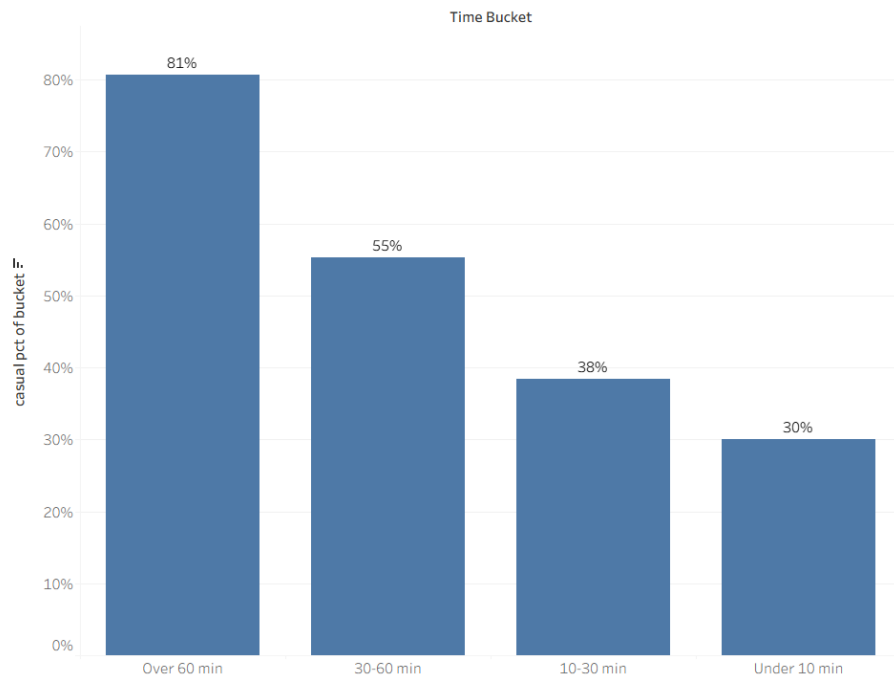
Key Findings

Finding 1. Ride duration

Casual riders average 19.1 minutes per ride (median 11.4) versus members' 12.0 minutes (median 8.4). Casual ride length is also nearly twice as variable (standard deviation ~39 vs. ~20), meaning casual riders mix quick trips with markedly longer ones more than members do.

Finding 2. Ride length distribution

Casual riders make up 35.96% of total rides, but that share climbs sharply with ride length. Casual riders account for 30% of rides under 10 minutes, 38% of rides between 10 and 30 minutes, 55% of rides between 30 and 60 minutes, and 81% of all rides over 60 minutes. Long rides are disproportionately casual.



Finding 3. Weekly pattern

Members peak Tuesday through Thursday and drop off on weekends, consistent with weekday commuting. Casual riders show the reverse pattern, peaking on Saturday, consistent with weekend leisure use.

Finding 4. Daily pattern

Members show two sharp ridership peaks, at 8am and 5pm, with a midday dip, classic commute timing. Casual riders show no morning peak at all, instead climbing gradually to a single afternoon peak at roughly half of members' peak volume.

Finding 5. Seasonality

Casual ridership is far more weather-dependent than member ridership. The casual-to-member ratio more than triples between winter and summer months, while member ridership stays comparatively steady year-round. This tracks with Chicago's seasons, spring is cold, windy, and rainy, while fall stays milder for longer, which fits casual ridership holding up better into fall than it does in spring.

Non-finding. Bike type

Classic bike versus electric bike usage shows no meaningful difference between casual and member riders (casual-to-member ratios of 0.52 for classic bikes and 0.58 for electric bikes). This was tested directly and is reported here as a pattern that does not hold, not omitted.

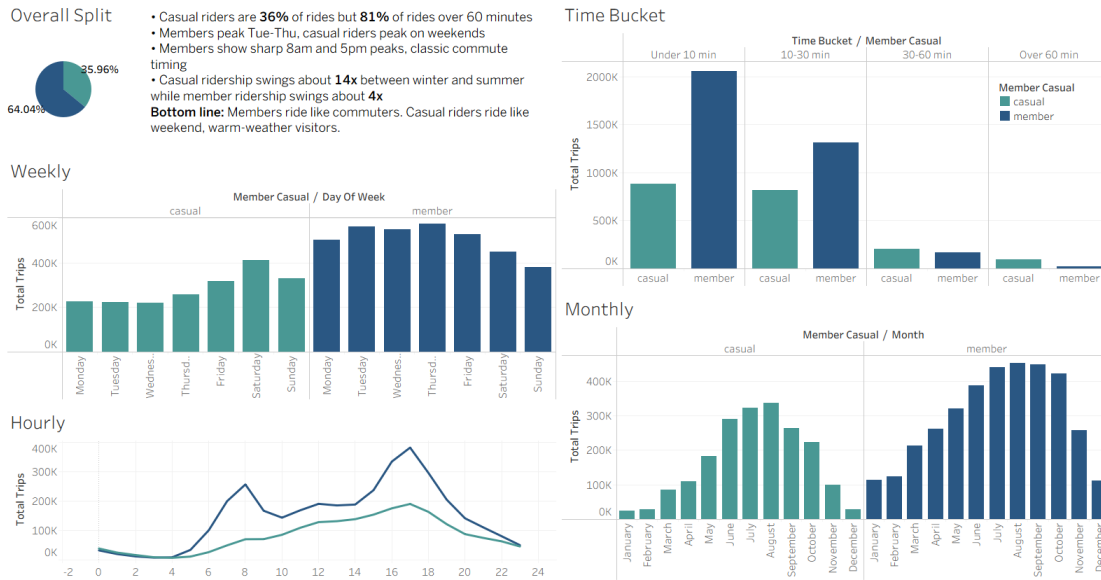
Non-finding. Geographic location

Mapping start and end coordinates shows both casual and member rides concentrated tightly around the lakefront and downtown corridor, with a sharp drop-off moving inland. Once member ridership's higher overall volume is accounted for, this does not represent a reliable geographic difference between the two groups and is not used as a basis for any recommendation below.

Interactive Dashboard

An interactive version of these findings, including a click-to-filter connection between the monthly and duration-bucket charts, is published on Tableau Public.

Casual Riders Ride Long, Members Ride Often



[View the live dashboard on Tableau Public](#)

Taken together, these findings describe a consistent story. Members use Cyclistic as reliable, short weekday transportation, while casual riders ride less often but disproportionately account for the longest, most leisure-oriented trips, concentrated on weekends and in warmer months.

III. Solutions

Three recommendations follow directly from the findings above, each with a genuine tradeoff rather than a one-sided pitch.

Recommendation	Pros	Cons
1. Long-ride membership tier	Directly targets the strongest, most quantified pattern in the data. Casual riders account for 80.64% of all rides over 60 minutes despite being only 35.96% of total ridership.	May cannibalize existing per-minute revenue from casual riders who take occasional long rides but were never going to convert to a full membership.
2. Weekend-specific membership promotion	Matches casual riders' actual usage pattern (Saturday peak) rather than pitching a generic commuter-style membership they don't need.	Weekend-only value framing could undersell the membership's weekday value or attract riders who churn once summer or weekend season ends.
3. May-through-October conversion campaign	Reaches casual riders when they are most numerous and most active, maximizing campaign reach per dollar spent.	A six-month campaign costs more and requires more marketing lead time than a shorter, single season push.

IV. Conclusion

This analysis confirms that annual members and casual riders use Cyclistic in fundamentally different ways, not just at different volumes, but with different rhythms, durations, and seasonal sensitivity. Members ride like commuters, taking short, frequent, weekday trips that hold steady across the year. Casual riders ride like

visitors and weekend users. Their rides are less frequent, but longer, more seasonally concentrated, and responsible for a disproportionate share of the platform's longest trips.

The clearest, most quantifiable opportunity for conversion lies in that duration gap. Eighty-one percent of hour-plus rides already belong to casual riders, meaning a membership built around long-ride value speaks directly to behavior these riders already exhibit, rather than asking them to adopt a commuter habit they don't have.

V. Next Steps

- **Pilot the long-ride membership tier.** Marketing team to design and price a membership tier emphasizing discounted or free time on rides over 30 minutes, targeted for pilot launch ahead of the summer season (recommended by May), with a 90-day performance review before a full rollout decision.
- **Launch the weekend promotion.** Marketing team to develop weekend-specific membership messaging and pricing, timed to launch alongside the long-ride pilot so both can be evaluated in the same season.
- **Schedule the seasonal campaign.** Marketing team to focus casual-conversion ad spend on May-October, with the heaviest weighting in June-August, using the pilot period's conversion data to guide the eventual full-year budget.
- **Further analysis.** Recommend a follow-up study using station-level or GPS-radius data (rather than raw lat/long density) to test whether casual rides genuinely cluster around specific tourist and lakefront corridors, this analysis found the map-based version of that claim inconclusive and did not include it above.